

Market Clearing with Complex Block Orders

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Nomenclature

Indices & Sets

- $i \in \mathcal{I}$ step orders;
- $j \in \mathcal{J}$ curtailable block orders;
- $l \in \mathcal{L}$ transmission lines;
- $m \in \mathcal{M}$ grid security constraints;
- $n \in \mathcal{N}$ nodes;
- $t \in \mathcal{T}$ time periods.

Parameters

- $Q_{i,t}$ order quantity (>0 : buy);
- $P_{i/j}$ order price;
- r_j minimum acceptance ratio;
- $a_{l,m}$ grid security incidence;
- $\beta_{n,i}$ flow incidence;
- $\gamma_{n,i}$ order incidence;
- $w_{m,t}$ grid security limit.

Variables

- $f_{l,t}$ power flow;
- $x_{i/j}$ order acceptance;
- y_j block order activation.

The primal market clearing problem

$$\max_{x_i, y_j, f_l} \left\{ \sum_{i \in \mathcal{I}} \left[\left(\sum_{t \in \mathcal{T}} Q_{i,t} \right) \cdot P_i \cdot x_i \right] + \sum_{j \in \mathcal{J}} \left(\sum_{t \in \mathcal{T}} Q_{j,t} \right) \cdot P_j \cdot x_j \right\} \quad (1)$$

s.t.:

$$x_i \leq 1, \quad \forall i \in \mathcal{I}, \quad [s_i] \quad (2)$$

$$y_j \leq 1, \quad \forall j \in \mathcal{J}, \quad [s_j] \quad (3)$$

$$y_j \cdot r_j \leq x_j \leq y_j, \quad \forall j \in \mathcal{J}, \quad [\underline{v}_j, \bar{v}_j] \quad (4)$$

$$\sum_{i \in \mathcal{I}} \gamma_{n,i} Q_{i,t} \cdot x_i + \sum_{j \in \mathcal{J}} \gamma_{n,j} Q_{j,t} \cdot x_j = \sum_{l \in \mathcal{L}} \beta_{n,l} \cdot f_{l,t}, \quad \forall (n, t) \in \mathcal{N} \times \mathcal{T}, \quad [\mu_{n,t}] \quad (5)$$

$$\sum_{l \in \mathcal{L}} a_{l,m} \cdot f_{l,t} \leq w_{m,t}, \quad \forall (m, t) \in \mathcal{M} \times \mathcal{T}, \quad [\lambda_{m,t}] \quad (6)$$

$$x_i, x_j, y_j \geq 0, \quad \forall i \in \mathcal{I}, \forall j \in \mathcal{J}, \quad (7)$$

$$y_j \in \mathbb{Z}, \quad \forall j \in \mathcal{J}. \quad (8)$$

Primal problem quick interpretation

- (1) Welfare function (primal variables);
- (2) upper bound for acceptance of stepwise orders;
- (3) upper bound for activation of block orders;
- (4) lower/upper bound for acceptance of curtailable block orders;
- (5) nodal power balance;
- (6) grid limits;
- (7) non-negativity of primal vars;
- (8) integrality of block order activation vars.

The dual market clearing problem

$$\min_{s_i, s_j, \mu_{n,t}, \lambda_{m,t}} \left\{ \sum_{i \in \mathcal{I}} s_i + \sum_{j \in \mathcal{J}} s_j + \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} w_{m,t} \cdot \lambda_{m,t} \right\} \quad (9)$$

s.t.:

$$s_i + \sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{T}} Q_{i,t} (\gamma_{n,i} \cdot \mu_{n,t}) - \sum_{t \in \mathcal{T}} Q_{i,t} \cdot P_i \geq 0, \quad \forall i \in \mathcal{I}, \quad [x_i] \quad (10)$$

$$\bar{v}_j - \underline{v}_j + \sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{T}} Q_{j,t} (\gamma_{n,j} \cdot \mu_{n,t}) - \sum_{t \in \mathcal{T}} Q_{j,t} \cdot P_j \geq 0, \quad \forall j \in \mathcal{J}, \quad [x_j] \quad (11)$$

$$s_j + r_j \cdot \underline{v}_j - \bar{v}_j \geq 0, \quad \forall j \in \mathcal{J}, \quad [y_j] \quad (12)$$

$$\sum_{m \in \mathcal{M}} a_{l,m} \cdot \lambda_{m,t} - \sum_{n \in \mathcal{N}} \beta_{n,l} \cdot \mu_{n,t} = 0, \quad \forall (l, t) \in \mathcal{L} \times \mathcal{T}, \quad [f_{l,t}] \quad (13)$$

$$s_i, s_j, \underline{v}_j, \bar{v}_j, \lambda_{m,t} \geq 0, \quad \forall q \in \mathcal{I} \cup \mathcal{J}, \quad \forall (m, t) \in \mathcal{M} \times \mathcal{T}. \quad (14)$$

Dual problem quick interpretation

- (9) Welfare function (dual variables);
- (10) dual constraint of stepwise order acceptance variable;
- (11) dual constraint of block order acceptance variable;
- (12) dual constraint of block order activation variable;
- (13) dual constraint of flow variable;
- (14) non-negativity of dual variables.

The complementary conditions

$$s_i \cdot (x_i - 1) = 0, \quad \forall i \in \mathcal{I}, \quad (15)$$

$$s_j \cdot (y_j - 1) = 0, \quad \forall j \in \mathcal{J}, \quad (16)$$

$$\underline{v}_j \cdot (r_j \cdot y_j - x_j) = 0, \quad \forall j \in \mathcal{J}, \quad (17)$$

$$\bar{v}_j \cdot (x_j - y_j) = 0, \quad \forall j \in \mathcal{J}, \quad (18)$$

$$x_i \cdot \left(s_i + \sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{T}} \mathcal{Q}_{i,t} (\gamma_{n,i} \cdot \mu_{n,t}) - \sum_{t \in \mathcal{T}} \mathcal{Q}_{i,t} \cdot P_i \right) = 0, \quad \forall i \in \mathcal{I}, \quad (19)$$

$$x_j \cdot \left(\bar{v}_j - \underline{v}_j + \sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{T}} \mathcal{Q}_{j,t} (\gamma_{n,j} \cdot \mu_{n,t}) - \sum_{t \in \mathcal{T}} \mathcal{Q}_{j,t} \cdot P_j \right) = 0, \quad \forall j \in \mathcal{J}, \quad (20)$$

$$y_j \cdot (s_j + r_j \cdot \underline{v}_j - \bar{v}_j) = 0, \quad \forall j \in \mathcal{J}, \quad (21)$$

$$\lambda_{m,t} \cdot \left(\sum_{l \in \mathcal{L}} a_{l,m} \cdot f_{l,t} - w_{m,t} \right) = 0, \quad \forall (m, t) \in \mathcal{M} \times \mathcal{T}. \quad (22)$$

Formulation of (relaxed) EEM – MPCC

Welfare maximization (1)

- ▶ upper bound for acceptance of stepwise orders (2);
- ▶ upper bound for activation of block orders (3);
- ▶ lower/upper bound for acceptance of curtailable block orders (4);
- ▶ nodal power balance (5);
- ▶ grid limits (6);
- ▶ non-negativity of primal vars (7);
- ▶ integrality of block order acceptance vars (8);

Formulation of (relaxed) EEM – MPCC

(continued)

- ▶ dual constraint of stepwise order acceptance variable (10);
- ▶ dual constraint of block order acceptance variable (11);
- ▶ dual constraint of block order activation variable (12);
- ▶ dual constraint of flow variable (13);
- ▶ non-negativity of dual variables (14);

Formulation of (relaxed) EEM – MPCC

(continued)

- ▶ complementarity constraints (15, 17 – 22);
- ▶ ~~complementarity between **block** order activation upper bound constraint & dual variable (16);~~
- ▶ price range constraints:

$$\mu_{\min} \leq \mu_{n,t} \leq \mu_{\max}, \quad \forall (n, t) \in \mathcal{N} \times \mathcal{T}. \quad (23)$$

Current solution approach for (relaxed) EEM – MPCC

- ▶ Complementarity.jl gets stuck in locally infeasible solutions!
- ▶ Using the classic Big-M reformulation:

$$u \cdot v = 0 \rightarrow \begin{cases} u \leq aux \cdot BigM \\ v \leq (1 - aux) \cdot BigM \\ aux \in \{0, 1\} \end{cases} \quad (24)$$

Formulation of equivalent EEM – MILP (1)

(1) Welfare maximization

- ▶ primal constraints (2 – 8);
- ▶ dual constraint of stepwise order acceptance variable (10);
- ▶ dual constraint of block order acceptance variable (11);
- ▶ dual constraint of flow variable (13);
- ▶ non-negativity of dual variables (14);
- ▶ price range constraints (23);

Formulation of equivalent EEM – MILP (1)

(continued)

- modified dual constraint of block order activation variable:

$$s_j + r_j \cdot \underline{v}_j - \bar{v}_j \geq -M_j \cdot (1 - y_j), \quad \forall j \in \mathcal{J}. \quad (25)$$

→ original constraint true only for accepted block orders.

- primal - dual objective equality:

$$\sum_{i \in \mathcal{I}} \left[\left(\sum_{t \in \mathcal{T}} Q_{i,t} \right) \cdot P_i \cdot x_i \right] + \sum_{j \in \mathcal{J}} \left(\sum_{t \in \mathcal{T}} Q_{j,t} \right) \cdot P_j \cdot x_j \geq \sum_{i \in \mathcal{I}} s_i + \sum_{j \in \mathcal{J}} s_j + \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} w_{m,t} \cdot \lambda_{m,t}. \quad (26)$$

Where $M_j := \sum_{t \in \mathcal{T}} 2 \cdot \mu^{\max} \cdot |Q_{j,t}| \quad \forall j \in \mathcal{J}$.

Paradoxical rejection example (2)

ID	Type	Quantity	Price	MAR
A	Step	10	300	–
B	Step	14	10	–
C	Block	-12	40	11/12
D	Step	-13	100	–

Paradoxical rejection example (2)

ID	<i>Primal</i>	<i>EEM – MPCC</i>	<i>EEM – MILP</i>
A	10	10	10
B	1	0	0
C	-11	0	0
D	0	-10	-10
Price	–	100	100
Σ Welf.	2570	2000	2000
Status	Paradoxical Acceptance	Paradoxical Rejection	

Introducing Exclusive Block Orders

Nomenclature – with exclusive block orders

Indices & Sets

$g \in \mathcal{G}_X$ exclusive block order groups;

$j_X \in \mathcal{J}_X$ blocks belonging in any exclusive group $g \in \mathcal{G}_X$.

Parameters

$e_{j,g}$ indicator taking the value 1 if block j belongs in group g .

The primal problem – with exclusive block orders

- Modified block order activation constraint (3):

$$y_j \leq 1, \quad \forall j \in \mathcal{J} \setminus \mathcal{J}_X, \quad [s_j] \quad (27)$$

$$\sum_{j \in \mathcal{J}} e_{j,g} y_j \leq 1, \quad \forall g \in \mathcal{G}_X. \quad [u_g] \quad (28)$$

CAUTION: I think that the constraint should be in the acceptance ratios!!

- **Required changes:** Modified block order activation constraint (3):

$$y_j \leq 1, \quad \forall j \in \mathcal{J}, \quad [s_j] \quad (29)$$

$$\sum_{j \in \mathcal{J}} e_{j,g} x_j \leq 1, \quad \forall g \in \mathcal{G}_X. \quad [u_g] \quad (30)$$

And onwards ...

The dual problem – with exclusive block orders

- Modified objective (9):

$$\min_{s_i, u_g, \mu_{n,t}, \lambda_{m,t}} \left\{ \sum_{i \in \mathcal{I}} s_i + \sum_{j \in \mathcal{J} \setminus \mathcal{J}_X} s_j + \sum_{g \in \mathcal{G}_X} u_g + \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} w_{m,t} \cdot \lambda_{m,t} \right\}; \quad (31)$$

- modified dual constraint of block order activation variable (12):

$$s_j + r_j \cdot \underline{v}_j - \bar{v}_j \geq 0, \quad \forall j \in \mathcal{J} \setminus \mathcal{J}_X, \quad [y_j] \quad (32)$$

$$\sum_{g \in \mathcal{G}_X} e_{j,g} \cdot u_g + r_j \cdot \underline{v}_j - \bar{v}_j \geq 0, \quad \forall j \in \mathcal{J}_X. \quad [y_j] \quad (33)$$

The complementarity conditions – with exclusive block orders

- Modified condition (16):

$$s_j \cdot (y_j - 1) = 0, \quad \forall j \in \mathcal{J} \setminus \mathcal{J}_X, \quad (\text{relaxed}) \quad (34)$$

- additional condition:

$$u_g \cdot \left(\sum_{j \in \mathcal{J}} e_{j,g} y_j - 1 \right) = 0, \quad \forall g \in \mathcal{G}_X, \quad (35)$$

- modified condition (21):

$$y_j \cdot (s_j + r_j \cdot \underline{v}_j - \bar{v}_j) = 0, \quad \forall j \in \mathcal{J} \setminus \mathcal{J}_X, \quad (36)$$

$$y_j \cdot \left(\sum_{g \in \mathcal{G}_X} e_{j,g} \cdot u_g + r_j \cdot \underline{v}_j - \bar{v}_j \right) = 0, \quad \forall j \in \mathcal{J}_X. \quad (37)$$

Formulation of EEM-MILP – with exclusive block orders

- Modified dual constraint of block order activation variable (25):

$$s_j + r_j \cdot \underline{v}_j - \bar{v}_j \geq -M_j \cdot (1 - \gamma_j), \quad \forall j \in \mathcal{J} \setminus \mathcal{J}_X, \quad (38)$$

$$\sum_{g \in \mathcal{G}_X} e_{j,g} \cdot u_g + r_j \cdot \underline{v}_j - \bar{v}_j \geq -M_j \cdot (1 - \gamma_j), \quad \forall j \in \mathcal{J}_X, \quad (39)$$

- modified primal - dual objective equality (26):

$$\sum_{i \in \mathcal{I}} \left[\left(\sum_{t \in \mathcal{T}} Q_{i,t} \right) \cdot P_i \cdot x_i \right] + \sum_{j \in \mathcal{J}} \left(\sum_{t \in \mathcal{T}} Q_{j,t} \right) \cdot P_j \cdot x_j \geq \sum_{i \in \mathcal{I}} s_i + \sum_{j \in \mathcal{J} \setminus \mathcal{J}_X} s_j + \sum_{g \in \mathcal{G}_X} u_g + \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} w_{m,t} \cdot \lambda_{m,t}. \quad (40)$$

Introducing Linked Block Orders

Nomenclature – with linked block orders

Indices & Sets

$j_p \in \mathcal{J}_p$ parent blocks,

$j_c \in \mathcal{J}_c(j_p)$ child blocks of parent block j_p ,

$j_c \in \mathcal{J}_c = \cup_{j_p \in \mathcal{J}_p} \mathcal{J}_c(j_p)$ all child blocks.

The primal problem – with linked block orders

- Modified block order activation constraints (3):

$$y_j \leq 1, \quad \forall j \in \mathcal{J} \setminus \mathcal{J}_c, \quad [s_j] \quad (41)$$

$$y_{j_c} \leq y_{j_p}, \quad \forall j_c \in \mathcal{J}_c(j_p), \quad \forall j_p \in \mathcal{J}_p. \quad [p_{j_c}] \quad (42)$$

The dual problem – with linked block orders

- Modified objective (9):

$$\min_{s_i, u_g, \mu_{n,t}, \lambda_{m,t}} \left\{ \sum_{i \in \mathcal{I}} s_i + \sum_{j \in \mathcal{J} \setminus \mathcal{J}_c} s_j + \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} w_{m,t} \cdot \lambda_{m,t} \right\}; \quad (43)$$

- modified dual constraint of block order activation variable (12):

$$s_j + r_j \cdot \underline{v}_j - \bar{v}_j \geq 0, \quad \forall j \in \mathcal{J} \setminus (\mathcal{J}_c \cup \mathcal{J}_p), \quad [y_j] \quad (44)$$

$$s_{j_p} - \sum_{j_c \in \mathcal{J}_c(j_p)} p_{j_c} + r_{j_p} \cdot \underline{v}_{j_p} - \bar{v}_{j_p} \geq 0, \quad \forall j_p \in \mathcal{J}_p, \quad [y_{j_p}] \quad (45)$$

$$p_{j_c} + r_{j_c} \cdot \underline{v}_{j_c} - \bar{v}_{j_c} \geq 0, \quad \forall j_c \in \mathcal{J}_c. \quad [y_{j_c}] \quad (46)$$

The complementarity conditions – with linked orders

- Modified condition (16):

$$s_j \cdot (y_j - 1) = 0, \quad \forall j \in \mathcal{J} \setminus \mathcal{J}_c, \quad (\text{relaxed}) \quad (47)$$

- additional condition:

$$p_{j_c} \cdot (y_{j_c} - y_{j_p}) = 0, \quad \forall j_c \in \mathcal{J}_c(j_p), \quad \forall j_p \in \mathcal{J}_p, \quad (48)$$

- modified condition (21):

$$y_j \cdot (s_j + r_j \cdot \underline{v}_j - \bar{v}_j) = 0, \quad \forall j \in \mathcal{J} \setminus (\mathcal{J}_c \cup \mathcal{J}_p), \quad (49)$$

$$y_{j_p} \cdot \left(s_{j_p} - \sum_{j_c \in \mathcal{J}_c(j_p)} p_{j_c} + r_{j_p} \cdot \underline{v}_{j_p} - \bar{v}_{j_p} \right) = 0, \quad \forall j_p \in \mathcal{J}_p, \quad (50)$$

$$y_{j_c} \cdot (p_{j_c} + r_{j_c} \cdot \underline{v}_{j_c} - \bar{v}_{j_c}) = 0, \quad \forall j_c \in \mathcal{J}_c, \quad (51)$$

Formulation of EEM-MILP – with linked block orders

- Modified dual constraint of block order activation variable (25):

$$s_j + r_j \cdot \underline{v}_j - \bar{v}_j \geq -M_j \cdot (1 - y_j), \quad \forall j \in \mathcal{J} \setminus (\mathcal{J}_c \cup \mathcal{J}_p), \quad (52)$$

$$s_{j_p} - \sum_{j_c \in \mathcal{J}_c(j_p)} p_{j_c} + r_{j_p} \cdot \underline{v}_{j_p} - \bar{v}_{j_p} \geq -M_j \cdot (1 - y_j), \quad \forall j_p \in \mathcal{J}_p, \quad (53)$$

$$p_{j_c} + r_{j_c} \cdot \underline{v}_{j_c} - \bar{v}_{j_c} \geq -M_j \cdot (1 - y_j), \quad \forall j_c \in \mathcal{J}_c. \quad (54)$$

- modified primal - dual objective equality (26):

$$\sum_{i \in \mathcal{I}} \left[\left(\sum_{t \in \mathcal{T}} Q_{i,t} \right) \cdot P_i \cdot x_i \right] + \sum_{j \in \mathcal{J}} \left(\sum_{t \in \mathcal{T}} Q_{j,t} \right) \cdot P_j \cdot x_j \geq \sum_{i \in \mathcal{I}} s_i + \sum_{j \in \mathcal{J} \setminus \mathcal{J}_c} s_j + \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} w_{m,t} \cdot \lambda_{m,t}. \quad (55)$$

More Examples

Base Case

ID	Zone	Qt1	Qt2	p0
1	1	100.0		300
2	2	50.0		300
3	3	40.0		300
4	1	-200.0		10
5	2	-100.0		20
6	3	-40.0		80
7	1		100.0	300
8	2		50.0	300
9	3		40.0	300
10	1		-200.0	10
11	2		-100.0	20
12	3		-40.0	80

ID	f12	f13	f23	Max(1)	Max(2)
11	1	1	0	80	
12	0	1	1		35
13	0	0	1		10
14	0	0	-1		60

ATC grid constraints

StepWise Orders

Base Case

109750.0	Zone1	Zone2	Zone3
Period 1	10.0	20.0	20.0
Period 2	10.0	10.0	80.0

Welfare & Zonal prices

	f12	f13	f23
Period 1	80.0	0.0	40.0
Period 2	-10.0	95.0	-60.0

Network flows

- ▶ Zone 1 (cheapest supply) export is limited by the grid at t1.
- ▶ Zone 3 (most expensive supply) import is limited by the grid at t2.

Introducing block orders

ID	Zone	Qt1	Qt2	p0
13	1	-20.0		5
14	3		-40.0	5

Block Orders

- Hint: cheap supply offers below the base-case market prices.

Introducing block orders

ID	Zone	Qt1	Qt2	p0	y	Welf.
13	1	-20.0		5	1.0	100.0
14	3		-40.0	5	1.0	200.0

Cleared Block Orders

110400.0	Zone1	Zone2	Zone3
Period 1	10.0	20.0	20.0
Period 2	10.0	10.0	10.0

Welfare & Zonal prices

	f12	f13	f23
Period 1	0.0	80.0	-40.0
Period 2	-10.0	60.0	-60.0

Network flows

- Zone 1 (cheapest supply) remains export-constrained at t1.

Declaring exclusive relationship

Declaring exclusive relationship

ID	Zone	Qt1	Qt2	p0	y	Welf.
13(EX1)	1	-20.0		5	0.0	0.0
14(EX1)	3		-40.0	5	1.0	200.0

Cleared Exclusive Block Orders

Declaring exclusive relationship

ID	Zone	Qt1	Qt2	p0	y	Welf.
13(EX1)	1	-20.0		5	0.0	0.0
14(EX1)	3		-40.0	5	1.0	200.0

Cleared Exclusive Block Orders

110300.0	Zone1	Zone2	Zone3
Period 1	10.0	20.0	20.0
Period 2	10.0	10.0	10.0

Welfare & Zonal prices

	f12	f13	f23
Period 1	0.0	80.0	-40.0
Period 2	-10.0	60.0	-60.0

Network flows

- We lost the welfare generated by order 13.

Introducing additional block orders

ID	Zone	Qt1	Qt2	p0
15	2	10.0		5
16	2		20.0	300
17	2		10.0	5

Input Block Orders

- Hint: Demand offers 15, 16 are below the current market prices.

Introducing additional block orders

ID	Zone	Qt1	Qt2	p0	y	Welf.
15	2	10.0		5	0.0	0.0
16	2		20.0	300	1.0	5800.0
17	2		10.0	5	0.0	0.0

Cleared Block Orders

116100.0	Zone1	Zone2	Zone3
Period 1	10.0	20.0	20.0
Period 2	10.0	10.0	10.0

Welfare & Zonal prices

	f12	f13	f23
Period 1	80.0	0.0	40.0
Period 2	10.0	60.0	-60.0

Network flows

- Only order 16 in the money.

Declaring parent-child relationships

Declaring parent-child relationships

ID	Role	Zone	Qt1	Qt2	p0	y	Welf.
15(FAM1)	parent	2	10.0		5	1.0	5650.0
16(FAM1)	child	2		20.0	300	1.0	–
17(FAM1)	child	2		10.0	5	0.0	–

Cleared Linked Block Orders

Declaring parent-child relationships

ID	Role	Zone	Qt1	Qt2	p0	y	Welf.
15(FAM1)	parent	2	10.0		5	1.0	5650.0
16(FAM1)	child	2		20.0	300	1.0	–
17(FAM1)	child	2		10.0	5	0.0	–

Cleared Linked Block Orders

115950.0	Zone1	Zone2	Zone3
Period 1	10.0	20.0	20.0
Period 2	10.0	10.0	10.0

Welfare & Zonal prices

	f12	f13	f23
Period 1	80.0	0.0	40.0
Period 2	10.0	60.0	-60.0

Network flows

- Order 16 saves its parent, but we lose some welfare.

Welfare evolution summary

Base Case	109750
+ Block Orders	110400
+ Exclusive Relationship	110300
+ Additional Block Orders	116100
+ Parent-child Relationship	115950

Welfare evolution

Price Evolution Summary

	Period 1			Period 2		
	Zn1	Zn2	Zn3	Zn1	Zn 2	Zn3
Base Case	10	20	20	10	10	80
+ Block Orders	10	20	20	10	10	10
+ Exclusive Relationship	10	20	20	10	10	10
+ Additional Block Orders	10	20	20	10	10	10
+ Parent-child Relationship	10	20	20	10	10	10

Price evolution

Base Case

ID	Zone	Qt1	Qt2	p0	x	Welf.
1	1	100.0		300	1.0	29000.0
2	2	50.0		300	1.0	14000.0
3	3	40.0		300	1.0	11200.0
4	1	-200.0		10	0.9	0.0
5	2	-100.0		20	0.1	0.0
6	3	-40.0		80	0.0	0.0
7	1		100.0	300	1.0	29000.0
8	2		50.0	300	1.0	14500.0
9	3		40.0	300	1.0	8800.0
10	1		-200.0	10	0.925	0.0
11	2		-100.0	20	0.0	0.0
12	3		-40.0	80	0.125	0.0

Cleared StepWise Orders

Block orders case

ID	Zone	Qt1	Qt2	p0	x	Welf.
1	1	100.0		300	1.0	29000.0
2	2	50.0		300	1.0	14000.0
3	3	40.0		300	1.0	11200.0
4	1	-200.0		10	0.8	0.0
5	2	-100.0		20	0.1	0.0
6	3	-40.0		80	0.0	0.0
7	1		100.0	300	1.0	29000.0
8	2		50.0	300	1.0	14500.0
9	3		40.0	300	1.0	11600.0
10	1		-200.0	10	0.75	0.0
11	2		-100.0	20	0.0	0.0
12	3		-40.0	80	0.0	0.0

Cleared StepWise Orders

Exclusive block orders case

ID	Zone	Qt1	Qt2	p0	x	Welf.
1	1	100.0		300	1.0	29000.0
2	2	50.0		300	1.0	14000.0
3	3	40.0		300	1.0	11200.0
4	1	-200.0		10	0.9	0.0
5	2	-100.0		20	0.1	0.0
6	3	-40.0		80	0.0	0.0
7	1		100.0	300	1.0	29000.0
8	2		50.0	300	1.0	14500.0
9	3		40.0	300	1.0	11600.0
10	1		-200.0	10	0.75	0.0
11	2		-100.0	20	0.0	0.0
12	3		-40.0	80	0.0	0.0

Cleared StepWise Orders

Additional block orders case

ID	Zone	Qt1	Qt2	p0	x	Welf.
1	1	100.0		300	1.0	29000.0
2	2	50.0		300	1.0	14000.0
3	3	40.0		300	1.0	11200.0
4	1	-200.0		10	0.9	0.0
5	2	-100.0		20	0.1	0.0
6	3	-40.0		80	0.0	0.0
7	1		100.0	300	1.0	29000.0
8	2		50.0	300	1.0	14500.0
9	3		40.0	300	1.0	11600.0
10	1		-200.0	10	0.85	0.0
11	2		-100.0	20	0.0	0.0
12	3		-40.0	80	0.0	0.0

Cleared StepWise Orders

Parent-child case

ID	Zone	Qt1	Qt2	p0	x	Welf.
1	1	100.0		300	1.0	29000.0
2	2	50.0		300	1.0	14000.0
3	3	40.0		300	1.0	11200.0
4	1	-200.0		10	0.9	0.0
5	2	-100.0		20	0.2	0.0
6	3	-40.0		80	0.0	0.0
7	1		100.0	300	1.0	29000.0
8	2		50.0	300	1.0	14500.0
9	3		40.0	300	1.0	11600.0
10	1		-200.0	10	0.85	0.0
11	2		-100.0	20	0.0	0.0
12	3		-40.0	80	0.0	0.0

Cleared StepWise Orders

References

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